THE COMPARISON DIAGRAMS OF THE QC AND ZARISKI TOPOLOGIES ON LOCALLY QUASI-COMPACT SPACES COMMUTE WITH A PROPER MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the identification of f_qc and the commutativity of the first diagram, the derivation of the second from the collected Zariski properties, and the reduction of the push-pull identity to part (7a) of those properties.

Editorial note. The label lemma-push-pull-LC names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the five cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The morphisms of topoi f_qc, epsilon, pi and the comparison a = pi o epsilon for LC_qc. Same chapter, the construction of the morphisms of topoi between the qc, Zariski and topological realisations of a morphism of locally quasi-compact spaces. Supplies every arrow of the two displayed diagrams.
- Relocalisation and the fibre-product description of the induced morphism. Sites, the lemmas describing the morphism of topoi induced by relocalisation and the continuous functor Z/Y |-> Z x_Y X/X. Supplies the identification of f_qc used in the proof.
- Collected properties of the Zariski comparison. Same chapter, the lemma collecting the compatibility statements for the Zariski topology, in particular parts (5) and (7a). Supplies the commutativity of the second diagram and the final identification pi_Y^{-1}f_*F = f_{Zar,*}pi_X^{-1}F.

Source reference. Original reference 0DCU: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7770–7770. Prerequisite pointer invoked at proof line(s) 7962; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 09X3: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 7675–7675. Prerequisite pointer invoked at proof line(s) 7969; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03CB: sources/stacks-chow/tags.txt, source lines 4378–4378. Prerequisite pointer invoked at proof line(s) 7960; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03EH: sources/stacks-chow/tags.txt, source lines 4456–4456. Prerequisite pointer invoked at proof line(s) 7954; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09W9: sources/stacks-chow/tags.txt, source lines 12487–12487. Prerequisite pointer invoked at proof line(s) 7957; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of LC_{qc}. Then there are commutative diagrams of topoi*

$$\begin{array}{ccc} Sh(LC_{qc}/X) & \xrightarrow{f_{qc}} & Sh(LC_{qc}/Y) \\ {\scriptstyle \epsilon_X}\downarrow & & \downarrow{\scriptstyle \epsilon_Y} \\ Sh(LC_{Zar}/X) & \xrightarrow{f_{Zar}} & Sh(LC_{Zar}/Y) \end{array} \quad \text{and} \quad \begin{array}{ccc} Sh(LC_{qc}/X) & \xrightarrow{f_{qc}} & Sh(LC_{qc}/Y) \\ {\scriptstyle a_X}\downarrow & & \downarrow{\scriptstyle a_Y} \\ Sh(X) & \xrightarrow{f} & Sh(Y) \end{array}$$

with $a_X = \pi_X \circ \epsilon_X$, $a_Y = \pi_X \circ \epsilon_X$. If f is proper, then $a_Y^{-1} \circ f_* = f_{qc,*} \circ a_X^{-1}$.

Proof. The morphism of topoi f_{qc} is the one from Sites, Lemma 03EH which in our case comes from the continuous functor $Z/Y \mapsto Z \times_Y X/X$, see Sites, Lemma 09W9. The diagram on the left commutes because the corresponding continuous functors compose correctly (see Sites, Lemma 03CB). The diagram on the right commutes because the one on the left does and because of part (5) of Lemma 0DCU.

Proof of the final assertion. The reader may repeat the proof of part (7a) of Lemma 0DCU; we will instead deduce this from it. As $\epsilon_{Y,*}$ is the identity functor on underlying presheaves, it reflects isomorphisms. The description in Lemma 09X3 shows that $\epsilon_{Y,*} \circ a_Y^{-1} = \pi_Y^{-1}$ and similarly for X. To show that the canonical map $a_Y^{-1} f_* \mathcal{F} \to f_{qc,*} a_X^{-1} \mathcal{F}$ is an isomorphism, it suffices to show that

$$\pi_Y^{-1} f_* \mathcal{F} = \epsilon_{Y,*} a_Y^{-1} f_* \mathcal{F} \to \epsilon_{Y,*} f_{qc,*} a_X^{-1} \mathcal{F} = f_{Zar,*} \epsilon_{X,*} a_X^{-1} \mathcal{F} = f_{Zar,*} \pi_X^{-1} \mathcal{F}$$

is an isomorphism. This is part (7a) of Lemma 0DCU. □